

Stochastic Processes - Stats 654 - Reference Sheet

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Problem-Solving Legend (What Tool Should I Try First?)

- **Step 0: Identify the object.**

- Is it about sums / random walks / hitting times? $\Rightarrow$ Think *martingales + optional stopping*.
- Is it about long-run behavior / stationary distribution / convergence? $\Rightarrow$ Think *Markov chains + mixing*.
- Is it about bounding $\mathbb{P}(\max \geq a)$ or tails? $\Rightarrow$ Think *Doob / Kolmogorov / Azuma / McDiarmid*.

- **Martingale tools (RW, hitting probs, expectations).**

- Try to build a martingale: S_n , $S_n - n\mu$, $S_n^2 - n\sigma^2$, ρ^{S_n} or a Doob martingale.
- If there is a natural random time (first hit, ruin, pattern), check if it is a stopping time and apply OST.

- **Markov chain tools (stationarity, convergence).**

- Compute π via $\pi P = \pi$ or symmetry/degree rule; check detailed balance for reversibility.
- Check irreducible + aperiodic; then use the convergence theorem or spectral gap/mixing bounds.

- **Coupling tools (mixing bounds, distances).**

- Try a shared-uniform coupling so both chains use the same randomness.
- Bound $\mathbb{P}(\tau > t)$ (meeting time) and then use $\|P^t(x, \cdot) - \pi\|_{\text{TV}} \leq \mathbb{P}(\tau > t)$.

- **Concentration tools (tail bounds).**

- Independent bounded variables $\Rightarrow$ Hoeffding / Chernoff.
- Martingale with bounded increments $\Rightarrow$ Azuma.
- Function of independent variables with bounded influence $\Rightarrow$ Doob martingale + McDiarmid.

- **MCMC tools (MH and Gibbs).**

- MH: write proposal Q , compute $\alpha(x, y) = \min\left(1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)}\right)$, then check detailed balance.
- Gibbs: derive full conditionals using the proportional trick, then argue invariance of π .

- **Graph / RW tools (recurrence, structure).**

- Random walk on graph: treat as Markov chain; stationary $\pi(x) \propto \deg(x)$ for undirected graphs.
- On trees or higher-dimensional lattices, reduce to a (possibly biased) 1D distance process and reuse gambler's ruin / biased RW results.

1 L1–L2: Conditional Expectation and Basic Martingales

Definition (Conditional Expectation). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, X an integrable random variable with $\mathbb{E}|X| < \infty$, and $\mathcal{G} \subseteq \mathcal{F}$ a sub- σ -algebra. A random variable Y is called the *conditional expectation* of X given $\mathcal{G}$ if:

1. Y is $\mathcal{G}$ -measurable,
2. for every $A \in \mathcal{G}$,

$$\int_A Y d\mathbb{P} = \int_A X d\mathbb{P}.$$

We write $Y = \mathbb{E}[X \mid \mathcal{G}]$. For a random variable Z , $\mathbb{E}[X \mid Z] := \mathbb{E}[X \mid \sigma(Z)]$.

Definition (Filtration). A *filtration* on $(\Omega, \mathcal{F}, \mathbb{P})$ is an increasing sequence of sub- σ -algebras

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}.$$

We interpret $\mathcal{F}_n$ as the information available up to time n .

Definition (Martingale, Submartingale, Supermartingale). Let $(\mathcal{F}_n)_{n \geq 0}$ be a filtration. A stochastic process $(M_n)_{n \geq 0}$ is a *martingale* w.r.t. $(\mathcal{F}_n)$ if:

1. M_n is $\mathcal{F}_n$ -measurable (adapted),
2. $\mathbb{E}|M_n| < \infty$ for every n ,
3. $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$ a.s. for all n .

It is a *submartingale* if $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \geq M_n$ a.s., and a *supermartingale* if $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \leq M_n$ a.s.

Example (Classic Martingales from Random Walks). Let $(X_i)_{i \geq 1}$ be i.i.d. with $\mathbb{E}X_i = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$, and define $S_n = \sum_{i=1}^n X_i$ with natural filtration $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$.

- If $\mu = 0$, then (S_n) is a martingale.
- $(S_n - n\mu)$ is always a martingale.
- $(S_n^2 - n\sigma^2)$ is a martingale.

- For simple symmetric random walk ($X_i = \pm 1$), S_n is a martingale and $S_n^2 - n$ is a martingale.

Definition (Stopping Time). Given a filtration $(\mathcal{F}_n)_{n \geq 0}$, a random variable $T : \Omega \rightarrow \{0, 1, 2, \dots\} \cup \{\infty\}$ is a *stopping time* if

$$\{T = n\} \in \mathcal{F}_n \quad \text{for all } n$$

equivalently, $\{T \leq n\} \in \mathcal{F}_n$ for all n . Intuitively, at time n we can decide whether T has occurred using only information up to n .

Definition (Stopped Process). Given a process (M_n) and stopping time T , the *stopped process* is

$$M_{n \wedge T} := M_{\min(n, T)}.$$

If (M_n) is a martingale, then $(M_{n \wedge T})$ is also a martingale.

2 L5–L7: Optional Stopping, Convergence, Transformations

Theorem 1 (Optional Stopping Theorem (Bounded Case)). *Let $(M_n)_{n \geq 0}$ be a martingale w.r.t. $(\mathcal{F}_n)$, and let T be a stopping time such that $T \leq N$ a.s. for some deterministic N . Then*

$$\mathbb{E}[M_T] = \mathbb{E}[M_0].$$

More generally, OST holds under suitable integrability and bounded-increment conditions.

Theorem 2 (Wald's Identity (Version I)). *Let $(X_i)_{i \geq 1}$ be i.i.d. with $\mathbb{E}|X_1| < \infty$ and mean μ , and let T be an integrable stopping time such that $\mathbb{E}T < \infty$ and certain regularity conditions hold (e.g. T bounded or suitable integrability). With $S_T = \sum_{i=1}^T X_i$, we have*

$$\mathbb{E}[S_T] = \mu \mathbb{E}[T].$$

This follows by applying OST to the martingale $S_n - n\mu$.

Definition (Almost Sure and L^1 Convergence). We say $X_n \rightarrow X$ almost surely (a.s.) if

$$\mathbb{P}(\{\omega : X_n(\omega) \rightarrow X(\omega)\}) = 1.$$

We say $X_n \rightarrow X$ in expectation (or in L^1) if $\mathbb{E}|X_n - X| \rightarrow 0$.

Theorem 3 (Monotone Convergence Theorem). *If (X_n) is a nondecreasing sequence of nonnegative random variables ($0 \leq X_1 \leq X_2 \leq \dots$) with $X_n \rightarrow X$ a.s., then*

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[X].$$

Theorem 4 (Dominated Convergence Theorem). *If $X_n \rightarrow X$ a.s. and there exists integrable Y with $|X_n| \leq Y$ a.s. for all n , then*

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[X].$$

Lemma 1 (Convex Transform of a Submartingale). *Let (Z_n) be a submartingale w.r.t. $(\mathcal{F}_n)$ and $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ convex, nondecreasing, with $\mathbb{E}|\varphi(Z_n)| < \infty$ for all n . Then $(\varphi(Z_n))$ is a submartingale.*

Theorem 5 (Not-To-Gamble Theorem). *Let (X_n) be a supermartingale and let (H_n) be a predictable sequence (i.e. H_n is $\mathcal{F}_{n-1}$ -measurable) with $H_n \geq 0$ a.s. Define the stochastic integral*

$$(H \cdot X)_n := \sum_{i=1}^n H_i(X_i - X_{i-1}), \quad (H \cdot X)_0 := 0.$$

Then $((H \cdot X)_n)$ is a supermartingale. Intuitively, no nonnegative predictable betting strategy can turn a losing game into a winning one in expectation.

Theorem 6 (Optional Stopping for Sub/Supermartingales (Bounded Case)). *Let (X_n) be adapted and T a stopping time with $T \leq N$ a.s.*

- *If (X_n) is a submartingale, then $\mathbb{E}[X_N] \geq \mathbb{E}[X_T] \geq \mathbb{E}[X_0]$.*
- *If (X_n) is a supermartingale, then $\mathbb{E}[X_N] \leq \mathbb{E}[X_T] \leq \mathbb{E}[X_0]$.*

3 L8–L11: Martingale Convergence and Inequalities

Theorem 7 (Convergence of Nonnegative Supermartingales). *If (X_n) is a supermartingale that is bounded below (e.g. $X_n \geq C$ a.s.), then there exists an integrable random variable X_∞ such that $X_n \rightarrow X_\infty$ almost surely and in L^1 under mild conditions.*

Definition (Doob's Maximal Inequality). *Let (M_n) be a nonnegative submartingale and $a > 0$. Then for any n ,*

$$\mathbb{P}\left(\max_{0 \leq k \leq n} M_k \geq a\right) \leq \frac{\mathbb{E}[M_n]}{a}.$$

This gives tail bounds for the running maximum of (M_n) .

Theorem 8 (Kolmogorov's Maximal Inequality). *Let $X_1, \dots, X_n$ be independent mean-zero random variables, and let $S_k = \sum_{i=1}^k X_i$. Then for any $a > 0$,*

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \geq a\right) \leq \frac{\text{Var}(S_n)}{a^2}.$$

Theorem 9 (Ville's Inequality). *Let (M_n) be a nonnegative martingale with $\mathbb{E}[M_0] = 1$. For any $a > 0$,*

$$\mathbb{P}\left(\sup_{n \geq 0} M_n \geq a\right) \leq \frac{1}{a}.$$

This is a uniform-in-time version of Markov's inequality for martingales.

Definition (Subgaussian Random Variable). A mean-zero random variable X is *subgaussian* with parameter σ^2 if

$$\mathbb{E}[\exp(\lambda X)] \leq \exp\left(\frac{\sigma^2 \lambda^2}{2}\right) \quad \forall \lambda \in \mathbb{R}.$$

Bounded variables are subgaussian with appropriate σ^2 .

Theorem 10 (Hoeffding's Inequality). *Let $X_1, \dots, X_n$ be independent with $a_i \leq X_i \leq b_i$ a.s., and define $S_n = \sum X_i$, $\mu = \mathbb{E}S_n$. Then for all $t > 0$,*

$$\mathbb{P}(S_n - \mu \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right),$$

and similarly for the lower tail. This is a concentration inequality for bounded independent sums.

Theorem 11 (Azuma–Hoeffding Inequality). *Let (M_n) be a martingale with $M_0 = 0$, and assume bounded differences:*

$$|M_n - M_{n-1}| \leq c_n \quad \text{a.s. for } n = 1, \dots, N.$$

Then for all $t > 0$,

$$\mathbb{P}(M_N \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{n=1}^N c_n^2}\right),$$

and similarly for $\mathbb{P}(M_N \leq -t)$. This is the martingale analogue of Hoeffding.

Theorem 12 (McDiarmid's Inequality). *Let $X_1, \dots, X_n$ be independent random variables taking values in some set $\mathcal{X}$, and let $f : \mathcal{X}^n \rightarrow \mathbb{R}$ satisfy a bounded differences condition:*

$$\sup_{x_1, \dots, x_n, x'_i} |f(x_1, \dots, x_i, \dots, x_n) - f(x_1, \dots, x'_i, \dots, x_n)| \leq c_i.$$

Let $Z = f(X_1, \dots, X_n)$ and $\mu = \mathbb{E}Z$. Then for all $t > 0$,

$$\mathbb{P}(Z - \mu \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right),$$

and similarly for $\mathbb{P}(Z - \mu \leq -t)$.

4 L12–L15: Markov Chains, Stationarity, Convergence

Definition (Discrete-Time Markov Chain). A stochastic process $(X_n)_{n \geq 0}$ with state space S is a *Markov chain* with transition matrix P if

$$\mathbb{P}(X_{n+1} = y \mid X_0, \dots, X_n) = \mathbb{P}(X_{n+1} = y \mid X_n) = P(X_n, y)$$

for all n and states y . The n -step transition probabilities are $P^n(x, y) := \mathbb{P}_x(X_n = y)$.

Definition (Stationary Distribution). A probability vector π on S is *stationary* for P if

$$\pi P = \pi.$$

Equivalently, if $X_0 \sim \pi$, then $X_n \sim \pi$ for all n .

Definition (Reversibility and Detailed Balance). A Markov chain with transition matrix P and distribution π is *reversible* w.r.t. π if

$$\pi(x)P(x, y) = \pi(y)P(y, x) \quad \forall x, y \in S.$$

If this holds, then π is stationary. This is called the detailed balance condition.

Definition (Irreducibility). A Markov chain with state space S is *irreducible* if for all $x, y \in S$ there exists $n \geq 1$ such that $P^n(x, y) > 0$. Intuitively, every state can reach every other state.

Definition (Period and Aperiodicity). The *period* of a state x is

$$d(x) := \gcd\{n \geq 1 : P^n(x, x) > 0\}.$$

If $d(x) = 1$, then x is aperiodic. In an irreducible chain, all states have the same period d . The chain is *aperiodic* if $d = 1$.

Theorem 13 (Existence and Uniqueness of Stationary Distribution (Finite Case)). *Every finite irreducible Markov chain has at least one stationary distribution. In the finite irreducible case this stationary distribution is unique.*

Definition (Total Variation Distance). For two probability measures μ, ν on a finite set,

$$\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_x |\mu(x) - \nu(x)| = \sup_A |\mu(A) - \nu(A)|.$$

Definition (Mixing Time). For a finite Markov chain with stationary distribution π , define

$$d(t) := \sup_x \|P^t(x, \cdot) - \pi\|_{\text{TV}}.$$

Then the *mixing time* is

$$t_{\text{mix}}(\varepsilon) := \min\{t : d(t) \leq \varepsilon\}.$$

Theorem 14 (Convergence Theorem). *For a finite, irreducible, aperiodic Markov chain with stationary distribution π ,*

$$\lim_{n \rightarrow \infty} \|P^n(x, \cdot) - \pi\|_{\text{TV}} = 0 \quad \text{for all } x.$$

Moreover π is unique.

5 L16–L18: Reversible Chains, Spectrum, Spectral Gap

Definition (Self-Adjointness and Spectrum for Reversible Chains). For a reversible finite-state Markov chain with stationary π , consider the inner product

$$\langle f, g \rangle_\pi := \sum_x f(x)g(x)\pi(x).$$

Then P is self-adjoint in $L^2(\pi)$:

$$\langle Pf, g \rangle_\pi = \langle f, Pg \rangle_\pi.$$

Hence all eigenvalues of P are real and lie in $[-1, 1]$, and there exists an orthonormal basis of eigenfunctions.

Definition (Spectral Gap and Relaxation Time). Let $1 = \lambda_1 > \lambda_2 \geq \dots \geq \lambda_n \geq -1$ be eigenvalues of P . Define

$$\lambda^* := \max_{j \geq 2} |\lambda_j|, \quad \gamma^* := 1 - \lambda^*, \quad t_{\text{rel}} := \frac{1}{\gamma^*}.$$

Here γ^* is the *absolute spectral gap* and t_{rel} is the *relaxation time*.

Theorem 15 (Spectral Bound on Mixing Time). *For a finite, irreducible, reversible Markov chain with stationary distribution π ,*

$$t_{\text{mix}}(\varepsilon) \leq t_{\text{rel}} \log \left(\frac{1}{\varepsilon \pi_{\min}} \right),$$

where $\pi_{\min} := \min_x \pi(x)$.

Example (Random Walk on an Undirected Graph). Let $G = (V, E)$ be a connected undirected graph. Simple random walk from x chooses a neighbor uniformly. Then

$$\pi(x) = \frac{\deg(x)}{2|E|}$$

is stationary and reversible. For regular graphs (e.g. cycles, hypercubes), π is uniform.

6 L19–L20: MCMC, Metropolis–Hastings, Gibbs

Definition (Metropolis–Hastings Algorithm). Let π be a target distribution on finite state space Ω , known up to a normalizing constant. Let $Q(x, y)$ be a proposal transition matrix. The MH chain has transition probabilities

$$P(x, y) = Q(x, y)\alpha(x, y) \quad (y \neq x),$$

where

$$\alpha(x, y) = \min \left(1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)} \right),$$

and

$$P(x, x) = 1 - \sum_{y \neq x} P(x, y).$$

Theorem 16 (Detailed Balance for MH). *The Metropolis–Hastings chain satisfies*

$$\pi(x)P(x, y) = \pi(y)P(y, x) \quad \forall x, y.$$

Hence π is stationary, and the chain is reversible w.r.t. π .

Definition (Full Conditional and Gibbs Sampler). Let π be a joint distribution on $\Omega = \Omega_1 \times \cdots \times \Omega_d$. The full conditional for coordinate i is

$$\pi(x_i \mid x_{-i}) := \mathbb{P}(X_i = x_i \mid X_j = x_j, j \neq i),$$

viewed as a distribution on Ω_i with other coordinates fixed.

Random-scan Gibbs sampler: at each step, choose coordinate i (e.g. uniformly) and resample X_i from $\pi(\cdot \mid X_{-i})$.

Lemma 2 (Proportional Trick for Conditionals). *If $\pi(x) \propto g(x)$ for some nonnegative function g , then*

$$\pi(x_i \mid x_{-i}) \propto g(x_i, x_{-i}) \quad \text{as a function of } x_i,$$

with normalization over $x_i \in \Omega_i$.

7 L21–L24: Coupling and Random Walk Recurrence

Definition (Coupling of Probability Measures). Let μ, ν be probability distributions on the same state space S . A *coupling* of (μ, ν) is a pair of random variables (X, Y) defined on a common probability space such that $X \sim \mu$ and $Y \sim \nu$.

Theorem 17 (Coupling Lemma). *For any two probability distributions μ, ν on a finite set S ,*

$$\|\mu - \nu\|_{\text{TV}} = \inf\{\mathbb{P}(X \neq Y) : (X, Y) \text{ is a coupling of } (\mu, \nu)\}.$$

Definition (Markovian and Faithful Coupling). Given a Markov chain with transition matrix P , a process (X_t, Y_t) on $S \times S$ is a *Markovian coupling* if each coordinate marginal evolves according to P :

$$\mathbb{P}(X_{t+1} = x' \mid X_t = x, Y_t = y) = P(x, x'), \quad \mathbb{P}(Y_{t+1} = y' \mid X_t = x, Y_t = y) = P(y, y').$$

The coupling is *faithful* if $X_t = Y_t$ implies $X_{t+1} = Y_{t+1}$ a.s. for all t .

Definition (Coupling Time and Coupling Inequality). For a coupling (X_t, Y_t) , the *coupling time* is

$$\tau := \inf\{t \geq 0 : X_t = Y_t\}.$$

For any initial states x, y ,

$$\|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}} \leq \mathbb{P}(X_t \neq Y_t) \leq \mathbb{P}(\tau > t).$$

In particular, if $Y_0 \sim \pi$, then $\|P^t(x, \cdot) - \pi\|_{\text{TV}} \leq \mathbb{P}(\tau > t)$.

Definition (Simple Random Walk on $\mathbb{Z}^d$). Let (X_n) be a Markov chain on $\mathbb{Z}^d$ where at each step the chain moves to a uniformly chosen neighbor (change one coordinate by ± 1). This is SRW on $\mathbb{Z}^d$.

Theorem 18 (Recurrence and Transience of SRW). *Simple random walk on $\mathbb{Z}^d$ is:*

- recurrent for $d = 1, 2$ (returns to the origin infinitely often with probability 1),
- transient for $d \geq 3$ (returns only finitely many times with positive probability).

Canonical Worked Examples

Example 1: Gambler's Ruin – Hitting Probability and Expected Time

Consider simple symmetric random walk on $\{0, 1, \dots, N\}$ with absorbing boundaries at 0 and N and $S_0 = i$.

(a) Probability of hitting N before 0. Let

$$p_i := \mathbb{P}_i(S_\tau = N), \quad \tau = \inf\{n \geq 0 : S_n \in \{0, N\}\}.$$

By the Markov property and symmetry,

$$p_i = \frac{1}{2}p_{i-1} + \frac{1}{2}p_{i+1}, \quad 1 \leq i \leq N-1,$$

with boundary conditions $p_0 = 0$, $p_N = 1$. This is a linear second-order difference equation with general solution $p_i = a + bi$. Using $p_0 = 0$ gives $a = 0$, and $p_N = 1$ gives $b = 1/N$. Thus

$$p_i = \frac{i}{N}.$$

(b) Expected time to absorption. Let

$$h_i := \mathbb{E}_i[\tau].$$

Again by conditioning on the first step,

$$h_i = 1 + \frac{1}{2}h_{i-1} + \frac{1}{2}h_{i+1}, \quad 1 \leq i \leq N-1,$$

with $h_0 = h_N = 0$. Rearranging:

$$h_{i+1} - 2h_i + h_{i-1} = -2.$$

A quadratic ansatz $h_i = \alpha i^2 + \beta i + \gamma$ yields

$$h_{i+1} - 2h_i + h_{i-1} = 2\alpha = -2 \Rightarrow \alpha = -1.$$

Write $h_i = -i^2 + \beta i + \gamma$. Using $h_0 = 0$ gives $\gamma = 0$, and $h_N = 0$ gives $-N^2 + \beta N = 0 \Rightarrow \beta = N$. Thus

$$h_i = i(N - i).$$

Example 2: Simple Random Walk Martingale + Azuma

Let (X_i) be i.i.d. with $\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}$ and $S_n = \sum_{i=1}^n X_i$.

Martingale. With $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$,

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + \mathbb{E}[X_{n+1}] = S_n,$$

so (S_n) is a martingale.

Azuma bound. Set $M_n = S_n$; then $|M_n - M_{n-1}| = |X_n| \leq 1$. Azuma–Hoeffding gives, for all $t > 0$,

$$\mathbb{P}(S_n \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{k=1}^n 1^2}\right) = \exp\left(-\frac{2t^2}{n}\right).$$

Similarly for $\mathbb{P}(|S_n| \geq t)$ by a union bound.

Example 3: Doob Martingale and McDiarmid

Let $X_1, \dots, X_n$ be independent and $f : \mathcal{X}^n \rightarrow \mathbb{R}$ satisfy the bounded differences condition

$$|f(x_1, \dots, x_i, \dots, x_n) - f(x_1, \dots, x'_i, \dots, x_n)| \leq c_i.$$

Define $Z = f(X_1, \dots, X_n)$ and the Doob martingale

$$M_k := \mathbb{E}[Z \mid X_1, \dots, X_k], \quad k = 0, \dots, n.$$

Then (M_k) is a martingale and one can check $|M_k - M_{k-1}| \leq c_k$ a.s. Applying Azuma,

$$\mathbb{P}(Z - \mathbb{E}Z \geq t) = \mathbb{P}(M_n - M_0 \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{k=1}^n c_k^2}\right),$$

which is McDiarmid's inequality.

Example 4: Random Walk on an Undirected Graph

Let $G = (V, E)$ be a finite connected undirected graph. Simple random walk: from x , move to a uniformly chosen neighbor.

Stationary distribution. Define

$$\pi(x) := \frac{\deg(x)}{2|E|}.$$

Then for any adjacent x, y ,

$$\pi(x)P(x, y) = \frac{\deg(x)}{2|E|} \cdot \frac{1}{\deg(x)} = \frac{1}{2|E|} = \frac{\deg(y)}{2|E|} \cdot \frac{1}{\deg(y)} = \pi(y)P(y, x),$$

and both sides are 0 if $(x, y) \notin E$. Thus detailed balance holds, so π is stationary and the chain is reversible.

Example 5: Coupling for a Two-State Chain

Consider $S = \{0, 1\}$ and

$$P = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix}.$$

Stationary distribution. Solve $\pi = \pi P$ with $\pi_0 + \pi_1 = 1$:

$$\pi_0 = \frac{3}{4}\pi_0 + \frac{1}{2}\pi_1 \Rightarrow \frac{1}{4}\pi_0 = \frac{1}{2}\pi_1 \Rightarrow \pi_1 = \frac{1}{2}\pi_0.$$

Normalization gives $\pi_0 + \frac{1}{2}\pi_0 = 1 \Rightarrow \pi_0 = 2/3, \pi_1 = 1/3$.

Coupling construction. Define $r(0) = 1/4, r(1) = 1/2$. Let $U_{t+1} \sim \text{Unif}(0, 1)$ i.i.d., and set

$$X_{t+1} = \mathbf{1}\{U_{t+1} \leq r(X_t)\}, \quad Y_{t+1} = \mathbf{1}\{U_{t+1} \leq r(Y_t)\}.$$

Each marginal has transition P , and if $X_t = Y_t$ then $X_{t+1} = Y_{t+1}$ (faithful).

If $X_t \neq Y_t$ then one is 0 and the other is 1. Under the Bernoulli coupling, the mismatch probability is

$$\mathbb{P}(X_{t+1} \neq Y_{t+1} \mid X_t \neq Y_t) = |r(1) - r(0)| = \frac{1}{4}.$$

Thus for $\tau = \inf\{t : X_t = Y_t\}$ and $X_0 \neq Y_0$,

$$\mathbb{P}(\tau > t) \leq \left(\frac{1}{4}\right)^t.$$

Mixing bound. Start $X_0 = x$ and $Y_0 \sim \pi$. Then $Y_t \sim \pi$ and

$$\|P^t(x, \cdot) - \pi\|_{\text{TV}} \leq \mathbb{P}(X_t \neq Y_t) \leq \mathbb{P}(\tau > t) \leq \left(\frac{1}{4}\right)^t.$$

Example 6: Lazy RW on Cycle – Order of Mixing

Lazy random walk on cycle of length ℓ : at each step, stay w.p. $1/2$, move to each neighbor w.p. $1/4$.

Stationary distribution. Underlying graph is regular; transition matrix is symmetric; hence π is uniform:

$$\pi(x) = 1/\ell.$$

Spectral gap and mixing. Non-lazy RW on cycle has eigenvalues $\lambda_k = \cos(2\pi k/\ell)$, so $1 - \lambda_2 \asymp 1/\ell^2$. Laziness maps eigenvalues to $\lambda'_k = \frac{1}{2}(1 + \lambda_k)$, so the absolute spectral gap remains $\gamma^* \asymp 1/\ell^2$, and $t_{\text{rel}} \asymp \ell^2$.

The spectral mixing bound gives

$$t_{\text{mix}}(\varepsilon) \lesssim t_{\text{rel}} \log\left(\frac{1}{\varepsilon \pi_{\min}}\right) \asymp \ell^2 \log(\ell/\varepsilon),$$

so for fixed ε , $t_{\text{mix}}(\varepsilon) = O(\ell^2 \log \ell)$.

Example 7: MH on $\{0, 1, 2, 3\}$ with Path Proposal

State space $\{0, 1, 2, 3\}$, target $\pi(i) \propto i + 1$ (weights $w(0) = 1, w(1) = 2, w(2) = 3, w(3) = 4$).

Proposal: random walk on the path

- from 0 propose 1;
- from 3 propose 2;
- from 1, 2 propose each neighbor w.p. 1/2.

Proposal kernel. Nonzero $q(i, j)$:

$$q(0, 1) = 1, \quad q(1, 0) = q(1, 2) = 1/2, \quad q(2, 1) = q(2, 3) = 1/2, \quad q(3, 2) = 1.$$

Acceptance probabilities. For neighbors,

$$\alpha(i, j) = \min \left(1, \frac{w(j)q(j, i)}{w(i)q(i, j)} \right).$$

Compute e.g. from 1:

$$\alpha(1, 0) = \min \left(1, \frac{1 \cdot 1}{2 \cdot (1/2)} \right) = 1, \quad \alpha(1, 2) = \min \left(1, \frac{3 \cdot (1/2)}{2 \cdot (1/2)} \right) = 1.$$

Thus

$$P(1, 0) = q(1, 0)\alpha(1, 0) = \frac{1}{2}, \quad P(1, 2) = q(1, 2)\alpha(1, 2) = \frac{1}{2}, \quad P(1, 1) = 0.$$

Similar calculations give all $P(i, j)$. Detailed balance with $\pi(i) \propto i + 1$ holds by MH construction.

Example 8: Gibbs for 2-Node Ising Model

Two spins $\sigma_1, \sigma_2 \in \{-1, +1\}$ with joint distribution

$$\pi(\sigma_1, \sigma_2) \propto \exp(\beta\sigma_1\sigma_2), \quad \beta > 0.$$

Unnormalized weights.

$$(+1, +1) : e^\beta; \quad (+1, -1) : e^{-\beta}; \quad (-1, +1) : e^{-\beta}; \quad (-1, -1) : e^\beta.$$

Full conditional for σ_1 . Using proportional trick,

$$\pi(\sigma_1 \mid \sigma_2) \propto \exp(\beta\sigma_1\sigma_2).$$

So if $\sigma_2 = +1$,

$$\mathbb{P}(\sigma_1 = +1 \mid \sigma_2 = +1) = \frac{e^\beta}{e^\beta + e^{-\beta}} = \frac{1}{1 + e^{-2\beta}},$$

and if $\sigma_2 = -1$,

$$\mathbb{P}(\sigma_1 = +1 \mid \sigma_2 = -1) = \frac{e^{-\beta}}{e^\beta + e^{-\beta}} = \frac{1}{1 + e^{2\beta}}.$$

Random-scan Gibbs step. At time t :

- With prob $1/2$, update σ_1 from $\pi(\cdot \mid \sigma_2)$.
- With prob $1/2$, update σ_2 from $\pi(\cdot \mid \sigma_1)$ (symmetric formula).

This Gibbs chain is reversible w.r.t. π and converges to π .

STAT 654 Cheatsheet (Key Facts)

Martingale: $(M_n, \mathcal{F}_n)$ with $\mathbb{E}|M_n| < \infty$ and $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$ a.s.; sub/supermartingale if $\geq / \leq$.

Stopping Time: T s.t. $\{T \leq n\} \in \mathcal{F}_n$ for all n ; decision to stop uses only past info.

Stopped Process: $M_{n \wedge T}$ is again a martingale if M_n is.

Optional Stopping (simple): If (M_n) martingale and T bounded a.s., then $\mathbb{E}[M_T] = \mathbb{E}[M_0]$.

Wald's Identity: i.i.d. X_i with mean μ , suitable stopping time T , then $\mathbb{E}\left[\sum_{i=1}^T X_i\right] = \mu \mathbb{E}[T]$.

Convex Transform: If (M_n) martingale and φ convex nondecreasing, then $(\varphi(M_n))$ is submartingale (when integrable).

Not-to-Gamble: If (X_n) supermartingale and $H_n \geq 0$ predictable, then $(\sum_{i=1}^n H_i(X_i - X_{i-1}))$ is supermartingale.

Supermartingale Convergence: Lower-bounded supermartingale converges a.s. (and in L^1 under mild conditions).

Doob Maximal Inequality: Nonnegative submartingale (M_k) , $a > 0$,

$$\mathbb{P}\left(\max_{0 \leq k \leq n} M_k \geq a\right) \leq \frac{\mathbb{E}[M_n]}{a}.$$

Kolmogorov Inequality: Independent mean-zero X_i , $S_k = \sum_{i=1}^k X_i$,

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \geq a\right) \leq \frac{\text{Var}(S_n)}{a^2}.$$

Ville Inequality: Nonnegative martingale (M_n) with $\mathbb{E}[M_0] = 1$,

$$\mathbb{P}\left(\sup_{n \geq 0} M_n \geq a\right) \leq \frac{1}{a}.$$

Hoeffding (bounded i.i.d.): $a_i \leq X_i \leq b_i$, $S_n = \sum X_i$, $\mu = \mathbb{E}S_n$,

$$\mathbb{P}(S_n - \mu \geq t) \leq \exp\left(-\frac{2t^2}{\sum (b_i - a_i)^2}\right).$$

Azuma–Hoeffding (martingale): $M_0 = 0$, $|M_n - M_{n-1}| \leq c_n$,

$$\mathbb{P}(M_N \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{n=1}^N c_n^2}\right).$$

McDiarmid (bounded differences): Independent X_i , f with change $\leq c_i$ per coordinate,

$$\mathbb{P}(f(X) - \mathbb{E}f(X) \geq t) \leq \exp\left(-\frac{2t^2}{\sum c_i^2}\right).$$

Gambler's Ruin (simple RW on $\{0, \dots, N\}$): Start at i , $\tau = \inf\{n : S_n \in \{0, N\}\}$,

$$\mathbb{P}_i(S_\tau = N) = \frac{i}{N}, \quad \mathbb{E}_i[\tau] = i(N - i).$$

SRW Recurrence: SRW on $\mathbb{Z}^d$ recurrent for $d = 1, 2$, transient for $d \geq 3$.

Markov Chain: $\mathbb{P}(X_{n+1} = y \mid X_0, \dots, X_n) = P(X_n, y)$; $P^n(x, y) = \mathbb{P}_x(X_n = y)$.

Stationary Distribution: π with $\pi P = \pi$; if $X_0 \sim \pi$ then $X_n \sim \pi$ for all n .

Reversible Chain: $\pi(x)P(x, y) = \pi(y)P(y, x)$ for all x, y (detailed balance), implies π stationary.

Irreducible: For all x, y , $\exists n : P^n(x, y) > 0$.

Period / Aperiodic: $d(x) = \gcd\{n \geq 1 : P^n(x, x) > 0\}$; chain aperiodic if $d(x) = 1$ for all x in class.

Convergence (finite, irr., aperiodic): Unique π , and $\|P^n(x, \cdot) - \pi\|_{\text{TV}} \rightarrow 0$ for all x .

Total Variation: $\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_x |\mu(x) - \nu(x)| = \sup_A |\mu(A) - \nu(A)|$.

Mixing Time: $d(t) = \sup_x \|P^t(x, \cdot) - \pi\|_{\text{TV}}$,

$$t_{\text{mix}}(\varepsilon) = \min\{t : d(t) \leq \varepsilon\}.$$

Coupling: (X, Y) with $X \sim \mu$, $Y \sim \nu$; $\|\mu - \nu\|_{\text{TV}} = \inf \mathbb{P}(X \neq Y)$.

Markovian Coupling: (X_t, Y_t) on S^2 where each marginal evolves by P ; faithful if $X_t = Y_t \Rightarrow X_{t+1} = Y_{t+1}$.

Coupling Time / Inequality: $\tau = \inf\{t : X_t = Y_t\}$,

$$\|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}} \leq \mathbb{P}(\tau > t).$$

Simple RW on Graph: From x , move to uniform neighbor; stationary $\pi(x) = \deg(x)/(2|E|)$ on connected undirected graphs.

Lazy RW: Stay w.p. $1/2$, otherwise move as original; kills period-2 issues on bipartite graphs.

Reversible Spectrum: P self-adjoint in $L^2(\pi)$; eigenvalues real in $[-1, 1]$.

Spectral Gap: $\lambda^* = \max_{j \geq 2} |\lambda_j|$, $\gamma^* = 1 - \lambda^*$, $t_{\text{rel}} = 1/\gamma^*$.

Spectral Mixing Bound:

$$t_{\text{mix}}(\varepsilon) \leq t_{\text{rel}} \log\left(\frac{1}{\varepsilon \pi_{\min}}\right), \quad \pi_{\min} = \min_x \pi(x).$$

Metropolis-Hastings: Proposal $Q(x, y)$, target π , accept prob

$$\alpha(x, y) = \min\left(1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)}\right).$$

Transition: $P(x, y) = Q(x, y)\alpha(x, y)$ for $y \neq x$, $P(x, x) = 1 - \sum_{y \neq x} P(x, y)$.

MH Detailed Balance: $\pi(x)P(x, y) = \pi(y)P(y, x)$, so MH chain is reversible w.r.t. π .

Gibbs Sampler: On $\Omega = \Omega_1 \times \dots \times \Omega_d$, update coordinate i by sampling from $\pi(x_i \mid x_{-i})$ (full conditional), leaving others fixed.

Proportional Trick: If $\pi(x) \propto g(x)$ then

$$\pi(x_i \mid x_{-i}) \propto g(x_i, x_{-i})$$

as a function of x_i (normalizing over x_i only).

Proof Techniques and Strategies (STAT 654)

2. Martingale Techniques

Checking Martingale Property: Given candidate (M_n) , compute $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n]$ explicitly. Use linearity, independence of increments, and that conditioning removes future randomness. Core examples: S_n , $S_n - n\mu$, $S_n^2 - n\sigma^2$, exponential martingales.

Constructing Martingales from Random Walks: Choose f so that $f(S_n)$ has zero conditional drift:

- Linear: S_n , $S_n - n\mu$.
- Quadratic: $S_n^2 - n\sigma^2$.
- Exponential: $r^{S_n} = \frac{q}{p}^{S_n}$ with r tuned to the step distribution (biased RW).

Then apply OST for hitting probabilities or expected times.

Designing Martingales for Hitting Problems: Pick parameters to make boundary values convenient. E.g. for biased RW on $\{0, \dots, N\}$, choose $M_n = \rho^{S_n}$ so that $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$, and M_τ has easily computable values at 0 and N .

Careful Use of Optional Stopping: When using OST:

- Show T is a stopping time.
- Check conditions: T bounded a.s., or integrability + bounded increments, etc.
- Write $\mathbb{E}[M_T] = \mathbb{E}[M_0]$ and expand M_T via terminal states (e.g. 0 vs N).

This is the template for gambler's ruin, biased RW, pattern waiting times.

Applying Maximal Inequalities: Identify whether you have:

- Nonnegative submartingale $\Rightarrow$ Doob maximal inequality.
- Independent mean-zero sums $\Rightarrow$ Kolmogorov inequality.
- Nonnegative martingale with $\mathbb{E}[M_0] = 1 \Rightarrow$ Ville inequality.

Verify assumptions, then plug into the inequality.

Doob Martingale + Azuma/McDiarmid: Given $Z = f(X_1, \dots, X_n)$ with independent X_i :

- Define $M_k = \mathbb{E}[Z \mid X_1, \dots, X_k]$.
- Show $|M_k - M_{k-1}| \leq c_k$ (bounded differences).
- Apply Azuma to (M_k) , which yields McDiarmid's inequality for f .

3. Markov Chain and Random Walk Techniques

Solving $\pi P = \pi$:

- Write linear equations for π and use $\sum_x \pi(x) = 1$.
- Exploit symmetry (e.g. regular graphs, symmetric transitions).
- Use degree rule: for simple RW on undirected graph, $\pi(x) \propto \deg(x)$.

Checking Detailed Balance (Reversibility): For candidate π ,

$$\pi(x)P(x, y) \stackrel{?}{=} \pi(y)P(y, x).$$

Check on edges / neighbor pairs; if holds and P is supported on edges, then π is stationary.

Irreducibility Tests: Show from any x you can reach any y with positive probability in some number of steps:

- For RW on graph: show graph is connected.
- For MCMC: show proposal and acceptance allow moving between all states.

Aperiodicity Tests:

- Show existence of self-loop $P(x, x) > 0$ in irreducible chain $\Rightarrow$ all states aperiodic.
- For cycles/paths: inspect possible return times to detect gcd of lengths.

Graph-RW as Markov Chain: Convert graph to P :

$$P(x, y) = \begin{cases} 1/\deg(x), & (x, y) \in E, \\ 0, & \text{otherwise.} \end{cases}$$

Then apply stationary/reversibility facts: $\pi(x) = \deg(x)/(2|E|)$.

Spectral Gap Reasoning: Use known behaviors:

- Cycle/path: $\text{gap} \asymp 1/n^2 \Rightarrow t_{\text{rel}} \asymp n^2$.
- Complete graph: gap bounded away from 0 $\Rightarrow t_{\text{rel}} = O(1)$.
- Hypercube: $\text{gap} \asymp 1/d$.

Then use

$$t_{\text{mix}}(\varepsilon) \lesssim t_{\text{rel}} \log \frac{1}{\varepsilon \pi_{\min}}.$$

4. Coupling Techniques

Shared-Randomness Markovian Coupling: Represent transitions from each state via intervals of $[0, 1]$:

- For each state i , partition $[0, 1]$ into subintervals of lengths $P(i, j)$.
- At time t , draw a single $U_{t+1} \sim \text{Unif}(0, 1)$ and update both X_{t+1} and Y_{t+1} using their respective partitions for X_t and Y_t .

This ensures each marginal evolves with kernel P .

Faithful Coupling Design: Ensure that if $X_t = Y_t$ then the transition rule from that state is the same for both, using the same randomness, so $X_{t+1} = Y_{t+1}$ a.s. and once they meet they stay together.

Geometric Tail Bounds for Coupling Time: If for all $i \neq j$,

$$\mathbb{P}(X_{t+1} \neq Y_{t+1} \mid X_t = i, Y_t = j) \leq c < 1,$$

then for $\tau = \inf\{t : X_t = Y_t\}$,

$$\mathbb{P}(\tau > t) \leq c^t.$$

Technique: compute or bound one-step “stay different” probability and iterate.

Distance-Based Couplings: Define a distance $d(X_t, Y_t)$ (graph distance, Hamming distance) and construct a coupling so d evolves as a simpler process:

- Show d has negative drift towards 0 or behaves like a biased RW hitting 0.
- Use hitting-time estimates on d to bound $\mathbb{P}(\tau > t)$.

Coupling with Stationary Initial State: To bound $\|P^t(x, \cdot) - \pi\|_{\text{TV}}$:

- Start $X_0 = x, Y_0 \sim \pi$.
- Run coupling so $Y_t \sim \pi$ for all t .
- Use $\|P^t(x, \cdot) - \pi\|_{\text{TV}} \leq \mathbb{P}(X_t \neq Y_t) \leq \mathbb{P}(\tau > t)$.

5. MCMC (MH and Gibbs) Techniques

Designing MH Proposals: Choose $Q(x, y)$ so moves are:

- Local (random-walk proposals on graph or path).
- Independence (always propose from a fixed distribution).
- Symmetric ($Q(x, y) = Q(y, x)$) to simplify acceptance ratios.

Computing MH Acceptances: Work with unnormalized target $w(x) \propto \pi(x)$:

$$\alpha(x, y) = \min \left(1, \frac{w(y)Q(y, x)}{w(x)Q(x, y)} \right).$$

Look for cancellations (especially if Q symmetric) and compute off-diagonal $P(x, y) = Q(x, y)\alpha(x, y)$, then diagonals by row sums.

MH Detailed Balance Proof: Show for $x \neq y$,

$$\pi(x)Q(x, y)\alpha(x, y) = \pi(y)Q(y, x)\alpha(y, x),$$

and note $P(x, x)$ defined by normalization gives detailed balance on diagonals as well.

Gibbs Full-Conditional Derivation (Proportional Trick): If $\pi(x) \propto g(x)$, then for coordinate i :

$$\pi(x_i \mid x_{-i}) \propto g(x_i, x_{-i}) \quad (\text{treat } x_{-i} \text{ as fixed constant}).$$

Normalize over $x_i \in \Omega_i$ only.

Gibbs Invariance/Stationarity: One-coordinate Gibbs update using full conditional $\pi(x_i \mid x_{-i})$ leaves π invariant; compositions of such updates (systematic or random scan) also leave π invariant, giving a reversible chain with stationary π .

6. Recurrence/Transience & Distance Processes

Reduction to Known SRW Results: If a process can be expressed as a function of a random walk:

- Map to SRW or biased RW on $\mathbb{Z}$ or $\mathbb{Z}_{\geq 0}$.
- Use known formulas (gambler's ruin, biased hitting probabilities) to infer recurrence, transience, or escape probabilities.

Distance-from-Root Processes: On trees or graphs, define $L_n = \text{dist}(X_n, \text{root})$:

- Show L_n is a Markov chain with probabilities “up 2/3, down 1/3” etc.
- Apply gambler's ruin or biased RW analysis to L_n .

7. General Inequality and Bounding Tactics

Union Bound: For events A_i ,

$$\mathbb{P}\left(\bigcup_i A_i\right) \leq \sum_i \mathbb{P}(A_i).$$

Used with tail bounds to control maxima over indices.

Jensen's Inequality: For convex φ and integrable X ,

$$\varphi(\mathbb{E}[X \mid \mathcal{G}]) \leq \mathbb{E}[\varphi(X) \mid \mathcal{G}].$$

Used in submartingale constructions and concentration proofs.

Norm/Distance Comparisons:

- Use coupling lemma: TV distance = inf disagreement probability over couplings.
- Use Cauchy–Schwarz in $L^2(\pi)$ and spectral decomposition for eigenvalue-based bounds.